METRIC

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DETAIL SPECIFICATION

LEAD PLATING, ELECTRODEPOSITED

This specification is approved for use by the Department of the Army and is available for use by all Departments and Agencies of the Department of Defense

1. SCOPE

1.1 Scope. This specification covers electrodeposited lead plating applied to iron and steel, copper, and copper-rich surfaces (see 6.1).

1.2 Classification. Lead plating should be of the following types and classes, as specified (see 6.2):

Type I - without preliminary copper coating

Class l - 25 micrometers (0.001 inch) thick lead coating
Class 2 - 13 micrometers (0.0005 inch) thick lead coating
Class 3 - 6 micrometers (0.00025 inch) thick lead coating

Type II - with 0.33 micrometer (0.000015 inch) thick preliminary copper coating (see 6.1.2)

Class l - 25 micrometers (0.001 inch) thick lead coating
Class 2 - 13 micrometers (0.0005 inch) thick lead coating
Class 3 - 6 micrometers (0.00025 inch) thick lead coating

Comments, suggestions, or questions on this document should be addressed to: Director, U.S. Army DEVCOM, Army Research Laboratory (ARL), Army Research Directorate, Specifications & Standards Office, Attn: FCDD-RLA-M, Aberdeen Proving Ground, MD 21005-5069. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil/.

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2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-1916	-	DoD Preferred Methods for Acceptance of Product Attributes

(Copies of these documents are available online at https://quicksearch.dla.mil/).

2.1.2 Other Government documents, drawings, and publications.

CODE OF FEDERAL REGULATIONS (CFR)

29 CFR 1910.1025	-	Toxic and Hazardous Substances (Lead)

(Copies of these documents are available online at https://www.osha.gov/).

2.2 Non-Government publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the contract or purchase order.

THE AMERICAN NATIONAL STANDARDS INSTITUTE

ANSI/ASQ Z1.4	-	Sampling Procedures and Tables for Inspection by Attributes
ISO 9000	-	Quality Management Systems - Fundamentals and Vocabulary

(Copies of this document are available online at https://ansi.org/).

ASTM INTERNATIONAL

ASTM B117	-	Standard Practice for Operating Salt Spray (Fog) Apparatus
ASTM B244	-	Standard Test Method for Measurement of Thickness of Anodic Coatings on Aluminum and of Other Nonconductive Coatings on Nonmagnetic Basis Metals with Eddy-Current Instruments
ASTM B487	-	Standard Test Method for Measurement of Metal and Oxide Coating Thickness by Microscopical Examination of Cross Section

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ASTM B499	-	Standard Test Method for Measurement of Coating Thicknesses by the Magnetic Method: Nonmagnetic Coatings on Magnetic Basis Metals
ASTM B567	-	Standard Test Method for Measurement of Coating Thickness by the Beta Backscatter Method
ASTM B568	-	Standard Test Method for Measurement of Coating Thickness by X-Ray Spectrometry

(Copies of these documents are available online at https://www.astm.org).

2.4 Order of precedence. Unless otherwise noted herein or within the contract or purchase order, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Materials. The materials used for electroplating shall produce coatings which meet the requirements of this specification.

3.2 Basis material. The basis material shall be free from defects that will be detrimental to the appearance or protective value of the plating.

3.3 General requirements.

3.3.1 Preplating operations. Unless otherwise specified in the contract or purchase order, the items to be plated shall be subjected to all the required mechanical operations, such as machining, brazing, welding, forming and perforating, prior to the application of the specified type and class of plating. The cleaning and finishing operations shall be performed with minimum abrasion, erosion, or pitting of the surfaces.

3.3.2 Stress relief treatment. All items made of metal having a hardness of Rockwell C40 or greater (see 3.3.3) shall be given a suitable heat treatment to remove any objectionable residual stresses, prior to being subjected to the final cleaning and plating operations.

3.3.3 Embrittlement relief. All items, such as springs, which are subject to flexure or repeated impact in their normal functions, or steels having a hardness of Rockwell C40 or greater, shall be heat treated for 3 hours at 190° ± 14 °C (375° ± 25 °F) after being plated. However, all hardened springs and carburized parts, which have been heat treated prior to plating at a temperature lower than 190 °C (375 °F) shall not be subjected to the specified after plating heat treatment. Instead, such parts shall be heat treated by a method or procedure which has the previous approval of the procuring agency. Plated springs shall not be flexed prior to the specified heat treatment in any case.

3.3.4 Deposition of the coating.

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3.3.4.1 Type I plating. The appropriate thickness of lead coating corresponding to the specified class of lead plating (see table I and 3.4.1.1), shall be electrodeposited directly on the surface to be protected.

3.3.4.2 Type II plating. A preliminary copper coating, of the thickness specified (see table II and 3.4.1.2), shall be electrodeposited directly on the surface to be protected. Subsequently the appropriate thickness of lead coating corresponding to the specified class of lead plating (see table I and 3.4.1.1), shall be electrodeposited on the copper coating.

3.4 Detail requirements.

3.4.1 Thickness of coating.

3.4.1.1 Type I plating. The average minimum thickness of the lead coating on all significant surfaces of the plated items (see 6.4) which can be touched by a ball 19 millimeters (0.75 inch) in diameter, shall be as shown in table I. The minimum thickness of the coating at any point on the surface of the plated item shall be not less than 70 percent of the minimum average thickness stipulated in table I for the specified class of plating. The thickness of the coating shall be determined by the applicable procedure specified in 4.4.1.

Table I. Thickness of lead coating for type I plating.

Class of Plating	Average Minimum Thickness	
	(Micrometers)	(Inches)
1	25	0.001
2	13	0.0005
3	6	0.00025

3.4.1.2 Type II plating. The average minimum thickness of the preliminary copper coating, and of the outer lead coating on all significant surfaces of the plated items (see 6.4) which can be touched by a ball 19 millimeters (0.75 inch) in diameter, shall be as shown in table II for the specified class of plating. The minimum thickness of the copper and lead coatings at any point on the surface of the plated item shall be not less than 70 percent of the minimum average thickness shown in table II for the specified class of plating. The thickness of the coatings shall be determined by the applicable procedure specified in 4.4.1.

Table II. Thickness of copper and lead coatings for type II plating.

Class of Plating	Average Minimum Thickness			
	Copper Coating		Lead Coating	
	(Micrometers)	(Inches)	(Micrometers)	(Inches)
1	0.33	0.000015	25	0.001
2	0.33	0.000015	13	0.0005
3	0.33	0.000015	6	0.00025

3.4.2 Salt spray (fog) requirements. When subjected to the 5 percent salt spray test specified in 4.4.2 for the applicable continuous length of time shown in table III for the specified type and class

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of plating, the coating on iron or steel basis metal shall give satisfactory protection against corrosion. For purposes of evaluating the protective value of the coating, the following test results shall be considered satisfactory:

(a) The formation of not more than 66 rust spots, none of which shall be larger than 1.6 millimeters (1/16 inch) in diameter, per square meter of plated surface (6 per square foot).

or:

(b) The formation of not more than 2 such rust spots when the coated surface area does not exceed 3 square decimeters (1/3 square foot).

The formation of any rust spot larger than 1.6 millimeter (1/16 inch) in diameter on any test specimen, regardless of total area of plated surface, or the occurrence of rust spots in excess of the limits specified above, shall be cause for rejection of the plating under test (see 4.4.2).

Table III. Salt spray (fog) exposure of type I or type II lead plating.

Type and Class of Plating	Salt Spray Exposure Time (Hours)
Class 1	96
Class 2	48
Class 3	24

3.5 Adhesion. The adhesion of the plating shall be such that when examined at a magnification of approximately 4 diameters, it shall not show separation from the basis metal at the interface when subjected to the test specified in 4.4.3. The interface between the lead and the basis metal is the surface of the basis metal before plating. The formation of cracks in the basis metal or plate which do not result in flaking, peeling, or blistering of the plate shall not be considered as nonconformance to this requirement.

3.6 Workmanship. The lead plating shall be smooth, fine grained, adherent, continuous, free from visible blisters, pits, modules, indications of burning, excessive build-up, staining and other defects. Superficial staining which has been demonstrated as resulting from rinsing, or slight discoloration resulting from baking operations to relieve embrittlement as specified above (see 3.3.3) shall not be cause for rejection. All details of workmanship shall conform to the best practice for high quality plating.

4. VERIFICATION

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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4.1.1 Quality assurance terms and definitions. Reference shall be made to ISO 9000 for definition of quality assurance terms.

4.2 Classification of inspection. The inspection requirements specified herein are classified as quality conformance inspection (see 4.3).

4.3 Quality Conformance Inspection.

4.3.1 Lot. A lot shall consist of plated articles of the same type and class and of approximately the same size, form and shape and plated under similar conditions, submitted for inspection at one time.

4.3.2 Sampling for visual examination and nondestructive tests. Sampling for visual examination and nondestructive tests shall be established in accordance with MIL-STD-1916 and conducted in accordance with ANSI/ASQ Z1.4. A sample of coated parts or articles shall be selected at random from each lot in accordance with ANSI/ASQ Z1.4, Level II. The lot shall be accepted or rejected according to the procedures in 4.3.2.1 for visual examination and 4.3.2.2 for plating thickness (nondestructive tests) at the Acceptable Quality Level (AQL) of 1.5 percent.

4.3.2.1 Visual examination. Samples selected in accordance with 4.3.2 shall be examined for compliance with the requirements of 3.6 after plating. If the number of nonconforming articles exceeds the acceptance number for the sample, the lot represented by the sample shall be rejected.

4.3.2.2 Thickness of plating (nondestructive tests). Samples selected in accordance with 4.3.2 shall be inspected and the plating thickness measured by the applicable tests detailed in 4.4.1. The part or article shall be considered nonconforming if one or more measurements fail to meet the minimum thickness requirements (see 3.4.1.1 and 3.4.1.2). If the number of defective items in any sample exceeds the acceptance number of the specified sample, the lot represented by the sample shall be rejected. Separate specimens (see 4.3.4) shall not be used for thickness measurements unless a need has been demonstrated. The number of separate specimens shall be as specified for sample in 4.3.3.

4.3.3 Sampling for destructive tests. A random sample selected in accordance with table IV, or separately plated specimens prepared in accordance with 4.3.4 shall be tested from each lot for each destructive test.

Table IV. Size of lot sample

Number of Items in the Lot	Number of Items Taken for Lot Sample
2 to 5	1
6 to 25	3
26 to 100	4
101 to 500	6
501 to 3200	10
Over 3200	1/

1/ For lot sizes over 3200 use the sampling schedule of ANSI/ASQ Z1.4, Level S-3, AQL 1.5 percent.

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4.3.3.1 Thickness of plating (destructive tests). Samples selected in accordance with 4.3.3 shall be measured for plating thickness by the applicable tests detailed in 4.4.1. The part or article shall be considered nonconforming if one or more measurements fail to meet the minimum thickness requirements (see 3.4.1.1 and 3.4.1.2). If the number of defective items in any sample exceeds the acceptance number of the specified sample, the lot represented by the sample shall be rejected. Separate specimens (see 4.3.4) shall not be used for thickness measurements unless a need has been demonstrated.

4.3.3.2 Adhesion (destructive tests). The articles or specimens used for the destructive thickness test (see 4.3.3.1), if of suitable size and form, may be used as the test pieces for the adhesion test to determine compliance with the requirements of 3.5. Failure of one or more of the test pieces shall constitute failure of the lot, except that for lot sizes over 3200 articles, the requirements of ANSI/ASQ Z1.4, inspection Level S-3, AQL 1.5 percent shall be followed. Separate specimens (see 4.3.4) may be used for the adhesion tests.

4.3.3.3 Salt spray (destructive tests). For both types I and II coatings, the number of articles indicated in table IV shall be taken at random from the lot to be inspected for the salt spray tests. Failure of one or more articles in the sample shall constitute failure of the lot, except that for lot sizes over 3200 articles, the requirements of ANSI/ASQ Z1.4, inspection Level S-3, AQL 1.5 percent shall be followed.

4.3.3.4 Embrittlement relief. For both types I and II coatings, the number of articles indicated in table IV shall be taken at random from the lot to be inspected for the embrittlement relief tests. Failure of one or more articles in the sample shall constitute failure of the lot, except that for lot sizes over 3200 articles, the requirements of ANSI/ASQ Z1.4, inspection Level S-3, AQL 1.5 percent shall be followed.

4.3.4 Separate specimens. When the plated articles are of such form as to be not readily adaptable to a test specified herein, the test may be made by the use of separate specimens plated concurrently with the articles represented. The separate specimens shall be of a basis metal equivalent to that of the articles represented. “Equivalent” basis metal includes condition and finish of surface prior to plating. For example, a smooth cold rolled surface should not be employed for representing a hot rolled surface, but due to the impracticability of hot forging or casting separate test specimens, hot rolled specimens may be used when plated parts are hot forged or cast. The separate specimens may be strips approximately 1 inch wide, 4 inches long and 0.04 inch thick, for adhesion tests, but shall be at least 4 inches wide, 6 inches long and approximately 0.04 inch thick for all other tests. These specimens shall be introduced into a lot at regular intervals prior to the cleaning operations preliminary to plating and shall not be separated therefrom until after completion of the processing. Conditions affecting the plating of the specimens including the spacing in respect to anodes and to other objects being plated shall correspond as nearly as possible to those affecting the significant surfaces of the articles represented.

4.4 Test methods.

4.4.1 Thickness. Separate specimens (see 4.3.4) shall not be used for thickness measurements unless a need for such specimens has been shown. Unless otherwise specified, the method of test shall be selected by the supplier with consideration given to basis metal, coating thickness, and the shape of the article. The following ASTM test methods have been found acceptable for measuring

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coating thickness: B244 (eddy-current), B487 (microscopic), B499 (magnetic), B567 (beta backscatter), and B568 (X-ray spectrometry).

4.4.2 Salt spray (fog) test. The salt spray test shall be conducted in accordance with ASTM B117.

4.4.3 Adhesion test. Wherever possible, the test shall be made on a specimen representing the basis material which can be bent 180°. The article or test specimen shall be clamped in a vise and the projecting portion bent back and forth until rupture occurs, or specimens shall be bent repeatedly over a mandrel until rupture occurs. Following fracture of the basis metal, it shall not be possible to detach areas of the coating with a sharp instrument.

4.4.3.1 Adhesion test surrogates. When the plated articles are not readily adaptable to the bend test, adhesion may be determined on the plated article or on a plated sample representative of the plated article (see 4.3.2) by cutting the plating from the basis metal at the interface in a continuous path. The plate shall then be examined at four diameters magnification to determine whether removal has been caused by the cutting away of an adherent plate or by the lifting of a nonadherent plate.

4.4.4 Embrittlement relief. Inspection for determination of compliance with 3.3.3 shall be made as specified by the procuring agency.

4.5 Retests. Plated items which have been rejected or withdrawn because of the presence of plating defects may be resubmitted after stripping and replating or after screening of the entire lot. Complete details of replating shall be furnished to, and be approved by, the procuring activity. Where only insufficient thickness is involved, replating without stripping may be permitted at the discretion of the procuring activity.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system command. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. Lead coatings are used to prevent corrosion, to prevent galling of metal parts, to improve the performance of bearings and to facilitate soldering.

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6.1.1 Type I. Lead may be plated directly on copper, brass or copper rich alloy surfaces. For steel surfaces, the type II process with a preliminary copper coating should be used to obtain adequate adhesion and corrosion resistance.

6.1.2 Type II. Type II should be used for all applications except where the presence of the initial thin copper coating is detrimental to the use. The thin undercoat of copper increases the salt spray resistance of lead coatings varying in thickness from 6 to 25 micrometers (0.00025 to 0.001 inch) by as much as a factor of two as well as improving adhesion of the lead to the steel. To attain 0.33 micrometer (0.000015 inch) of copper from a cyanide bath operating at 50 percent cathode efficiency will require about l minute of plating at a current density of 16 amperes per square foot or correspondingly shorter period for higher current densities.

6.1.3 Health and safety considerations. Lead can cause birth defects or other reproductive harm. Exposure to lead during pregnancy can affect brain development and cause learning and behavior problems for the child. Exposure to lead can harm the reproductive systems of men and women. Exposure to lead and lead compounds may also increase cancer risk. For these reasons, the inclusion of lead on DoD equipment is often restricted by contractual requirements. Lead has been listed in 29 CFR 1910 (OSHA Standards) under “Toxic and Hazardous Substances” and personnel exposure to these materials during the process must be limited to the values specified in the applicable portions of the OSHA Standard 1910.1025. All applicable safety rules, regulations, and procedures should be followed during the inspection, sampling, and testing of this material.

6.2 Ordering data. Purchasers should exercise any option offered herein and procurement documents should specify the following:

(a) Title, number and date of this specification.

(b) Type and class of plating required (see 1.2).

(c) Method of embrittlement relief testing (see 4.4.4).

6.3 Plating facility data. The manufacturer of the basis metal parts should provide the plating facility with the following data:

(a) Hardness of steel parts (see 3.3.2 and 3.3.3).

(b) Whether heat treatment for stress relief has been performed or is required.

(c) Ultimate use (see 3.3.3).

(d) Tensile loads and time required for embrittlement relief test (if applicable see 4.4.4).

6.4 Significant surfaces. Significant surfaces (3.4.1.1) are those surfaces that are visible and subject to wear and corrosion.

6.5 Subject term (key word) listing.

Circuit Boards Electronics Soldering Wear

6.6 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

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Review activities:
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